

Setup system

Setup the system and get files from the internet.

In the following code, we will download the necessary files from the internet.

```

1  # SETUP: Please ignore this part.
2
3  # Remove existing files
4  import os
5
6  # Stop pygame from opening a window
7  os.environ["SDL_VIDEODRIVER"] = "dummy"
8
9  # Install kaleido package
10 !pip install -U kaleido==0.2.1
11
12 # Mount the drive:
13 from google.colab import drive
14 drive.mount('/content/drive')
15
16 try:
17     os.chdir("/content/drive/MyDrive/Race-2025")
18     print("Directory changed successfully.")
19 except FileNotFoundError:
20     print("Error: Race-2025 directory not found.")
21
22 # Create a local directory
23 !mkdir /content/drive/MyDrive/Race-2025
24
25 # Get the repo as a zip file and save it locally:
26 !wget https://github.com/pattichis/GraphSpeeds2/archive/refs/heads/main.zip -O repo.zip
27 !unzip repo.zip -d /content/drive/MyDrive/Race-2025/
28 !rm -f repo.zip
29
30
31 #!find './' -maxdepth 1 -type f -delete
32
33 # Use wget with -nc to avoid overwriting local copy.
34 !rm -f lineart_v4.py*
35 !wget https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/lineart_v4.py
36
37 # Download the pictures:
38 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/Elephant.png
39 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/Squirrel.png
40 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/Tortoise.jpg
41 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/Koala.jpeg
42
43 # Import the functions to use
44 from lineart_v4 import table, simulationVideo, textImage, plotTablesLines, CreateVideo, changeVideoSpeed

```

Collecting kaleido==0.2.1

Downloading kaleido-0.2.1-py2.py3-none-manylinux1_x86_64.whl.metadata (15 kB)

Downloading kaleido-0.2.1-py2.py3-none-manylinux1_x86_64.whl (79.9 MB)

79.9/79.9 MB 9.3 MB/s eta 0:00:00

Installing collected packages: kaleido

Successfully installed kaleido-0.2.1

Mounted at /content/drive

Directory changed successfully.

--2026-08-20 17:37:16-- https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/lineart_v4.py

Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.110.133, 185.199.108.13

Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.

HTTP request sent, awaiting response... 200 OK

Length: 34249 (33K) [text/plain]

Saving to: 'lineart_v4.py'

lineart_v4.py 100%[=====>] 33.45K --.-KB/s in 0.003s

2026-08-20 17:37:16 (10.3 MB/s) - 'lineart_v4.py' saved [34249/34249]

File 'Elephant.png' already there; not retrieving.

--2026-08-20 17:37:16-- <https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/Squirrel.png>

Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.111.133, 185.199.110.13

```

Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-08-20 17:37:16 ERROR 404: Not Found.

File 'Tortoise.jpg' already there; not retrieving.

File 'Koala.jpeg' already there; not retrieving.

pygame 2.6.1 (SDL 2.28.4, Python 3.12.13)
Hello from the pygame community. https://www.pygame.org/contribute.html
/usr/local/lib/python3.12/dist-packages/moviepy/config_defaults.py:47: SyntaxWarning: invalid escape sequence '\P'
  IMAGEMAGICK_BINARY = r"C:\Program Files\ImageMagick-6.8.8-Q16\magick.exe"
/usr/local/lib/python3.12/dist-packages/moviepy/video/io/ffmpeg_reader.py:294: SyntaxWarning: invalid escape sequen
  lines_video = [l for l in lines if ' Video: ' in l and re.search('\d+x\d+', l)]
/usr/local/lib/python3.12/dist-packages/moviepy/video/io/ffmpeg_reader.py:367: SyntaxWarning: invalid escape sequen
  rotation_lines = [l for l in lines if 'rotate' : ' in l and re.search('\d+$', l)]
/usr/local/lib/python3.12/dist-packages/moviepy/video/io/ffmpeg_reader.py:370: SyntaxWarning: invalid escape sequen
  match = re.search('\d+$', rotation_line)
WARNING:py.warnings:/usr/local/lib/python3.12/dist-packages/moviepy/video/io/sliders.py:61: SyntaxWarning: "is" wit
  if event.key is 'enter':

```

Build Tables

In coding, objects are used to save functions and their data for processing. We can create an object using a constructor.

Here we create a table object to store information using:

```

# Enter the table values:
column_labels = ["x (hours)", "y (miles)"]
time_values = [0, 1, 3, 4, -1] # <- Replace -1 value by correct one.
distance_values = [0, 11, 33, -1, 88] # <- Replace -1 value by correct one.

# Create a table for the Squirrel data
Squirrel = table(x=time_values, y=distance_values, column_labels=column_labels)

```

We can see what we stored using `print()`.

However, a nicer way to see the values is to use `plotTable()`:

```
Squirrel.plotTable()
```

You can save an image using `saveImage()` as shown below:

```
Squirrel.saveImage("squirrel-table.jpg") # <- Can you find this image?
```

This will save the table in the file `squirrel-table.png`.

Assignment: Fill in the tables

Study the code below. Then perform the following:

1. Can you identify a common rate or speed between time and distance?
2. Replace the -1 values with the correct values. Discuss within your group.
3. Click on the generated table image and open it.
4. What is the filename of each table image?

```

1 # Enter the table values:
2 column_labels = ["x (hours)", "y (miles)"]
3 time_values = [0, 1, 3, 4, 8] # <- Replace -1 value by correct one.
4 distance_values = [0, 11, 33, 44, 88] # <- Replace -1 value by correct one.
5
6 # Create a table for the Squirrel data
7 Squirrel = table(x=time_values, y=distance_values, column_labels=column_labels)
8
9 # Plot the table
10 Squirrel.plotTable()
11

```

```

12 # Save the image for the table
13 Squirrel.saveImage("squirrel-table.jpg") # <- Can you find this image?

```

x (hours)	y (miles)
0.00000	0.00000
1.00000	11.00000
3.00000	33.00000
4.00000	44.00000
8.00000	88.00000

Wrote squirrel-table.jpg

```

1 # Enter the table values:
2 column_labels = ["x (hours)", "y (miles)"]
3 time_values = [0, 3, 5, 6, 10] # <- Replace -1 values with correct ones.
4 distance_values = [0, 18, 30, 36, 60] # <- Replace -1 values with correct ones.
5
6 # Create a table for the Elephant data
7 Elephant = table(x=time_values, y=distance_values, column_labels=column_labels)
8
9 # Plot the table
10 Elephant.plotTable()
11
12 # Save the image for the table
13 Elephant.saveImage("elephant-table.jpg") # <- Can you find this image?

```

x (hours)	y (miles)
0.00000	0.00000
3.00000	18.00000
5.00000	30.00000
6.00000	36.00000
10.00000	60.00000

Wrote elephant-table.jpg

```

1 # Enter the table values:
2 column_labels = ["x (hours)", "y (miles)"]
3 time_values    = [0, 3, 5, 8, 12] # <- Change -1 to correct value.
4 distance_values = [0, 3.6, 6, 9.6, 14.4] # <- Change -1 to correct value.
5 data_values = [time_values, distance_values]
6
7 # Create the table:
8 Tortoise = table(x=time_values, y=distance_values, column_labels=column_labels)
9
10 # Plot the table
11 Tortoise.plotTable()
12
13 # Save the image for the table
14 Tortoise.saveImage("tortoise-table.jpg") # <- Find this image

```

x (hours)	y (miles)
0.00000	0.00000
3.00000	3.60000
5.00000	6.00000
8.00000	9.60000
12.00000	14.40000

Wrote tortoise-table.jpg

```

1 # Enter the table values:
2 column_labels = ["x (hours)", "y (miles)"]
3 time_values    = [0, 1, 3, 5, 6] # Change -1 to the correct value.
4 distance_values = [0, 6.2, 18.6, 31, 37.2] # Change -1 to the correct value.
5 data_values = [time_values, distance_values]
6
7 # Create the table:
8 Koala = table(x=time_values, y=distance_values, column_labels=column_labels)
9
10 # Plot the table
11 Koala.plotTable()
12
13 # Save the image for the table
14 Koala.saveImage("koala-table.jpg") # <- Find this image.

```

x (hours)	y (miles)
0.00000	0.00000
1.00000	6.20000
3.00000	18.60000
5.00000	31.00000
6.00000	37.20000

Wrote koala-table.jpg

Enter the equations

Assignment: Enter the equations and plot the graph

1. Discuss the code within your group.
2. Correct the equations $y=?*x$ for each animal.
3. Move your cursor over points in the graph. Where do these points come from?
4. Discuss the slope of each line. How do they correspond to the equations?

Modify or add to your answers:

1. Who is going the fastest? How can you tell from the equations?
2. Who is going the slowest? How can you tell from the graphs?
3. How do the tables help support these answers?

```

1 # Merge all of the tables
2 tables_list = [Elephant, Squirrel, Tortoise, Koala]
3
4 # Line equations of tables
5 legend_labels = [
6     'Elephant: y = 6*x', # <- Modify the equation here.
7     'Squirrel: y = 11*x', # <- Modify the equation here.
8     'Tortoise: y = 1.2*x', # <- Modify the equation here.
9     'Koala: y = 6.2*x'] # <- Modify the equation here.
10
11 # Plot tables
12 plotTablesLines(tables_list,
13                 fig_title = 'Compare speeds', # title on top
14                 x_label = 'Elapsed time (hours)', # appears along x-axis
15                 y_label = 'Distance in miles', # appears along y-axis
16                 legend_title = 'Journey', # title on the right hand side
17                 legend_labels = legend_labels, # text describing the table
18                 img_name = 'plot.jpg') # plot image name

```



Make videos I: Pick Find character images

Assignment: Find character images

1. Click on your Folder icon on the Left Hand Side.
2. Click on drive.
3. Click on MyDrive.
4. Click on Race-2025.
5. Click on GraphSpeeds2-main.
6. Look at the different images.
7. Write down the names of the image files that you want to use.

✖ Make videos II: Create tables for your characters

✖ Assignment: Create the tables and equations for your characters

1. Enter time and distance units in column_labels.
2. Enter time_values and distance_values for each value.
3. Enter names and equations for each character.

```

1 # Prepare the labels for everyone
2 column_labels = ["x (seconds)", "y (feet)"] # <- FIX units
3
4 # Table #1 values:
5 time_values    = [0, 1, 2, 4] # <- ENTER x values
6 distance_values = [0, 12.5, 25, 50] # <- ENTER y values
7
8 # Create your table:
9 char1 = table(x=time_values, y=distance_values, column_labels=column_labels)
10 char1.plotTable()
11 char1.saveImage("char1.png")
12
13
14
15 # Table #2 values:
16 time_values    = [0, 1, 2, 3.5, 7] # <- ENTER x values

```

```

17 distance_values = [0,7.14,14.28,25,50] # <- ENTER y values
18
19 # Create your table:
20 char2 = table(x=time_values, y=distance_values, column_labels=column_labels)
21 char2.plotTable()
22 char2.saveImage("char2.png")
23
24
25
26 # Table #3 values:
27 time_values      = [0,1,2,6.51,13] # <- ENTER x values
28 distance_values = [0, 3.84,7.68,25,50] # <- ENTER y values
29
30 # Create your table:
31 char3 = table(x=time_values, y=distance_values, column_labels=column_labels)
32 char3.plotTable()
33 char3.saveImage("char3.png")
34
35
36
37 # Table #4 values:
38 time_values      = [0, 1,2,4,8] # <- ENTER x values
39 distance_values = [0, 6.25,12.5,25,50] # <- ENTER y values
40
41 # Create your table:
42 char4 = table(x=time_values, y=distance_values, column_labels=column_labels)
43 char4.plotTable()
44 char4.saveImage("char4.png")
45
46
47 # Merge all of the tables
48 tables_list = [char1, char2, char3, char4]
49
50 # Line equations of tables
51 legend_labels = [
52     'race car: y = 12.5*x', # <- Modify name and the equation here.
53     'doggy: y = 7.14*x', # <- Modify name and the equation here.
54     'curious george: y = 3.84*x', # <- Modify name and the equation here.
55     'low-bike: y = 6.25*x'] # <- Modify name and the equation here.
56
57 # Plot tables
58 plotTablesLines(tables_list,
59                 fig_title = 'The Race', # title on top
60                 x_label = 'Elapsed time (hours)', # appears along x-axis
61                 y_label = 'Distance in feet', # appears along y-axis
62                 legend_title = 'Journey', # title on the right hand side
63                 legend_labels = legend_labels, # text describing the table
64                 img_name = 'race.png') # Saves under race.png.

```

x (seconds)	y (feet)
0.00000	0.00000
1.00000	12.50000
2.00000	25.00000
4.00000	50.00000

Wrote char1.png

x (seconds)	y (feet)
0.00000	0.00000
1.00000	7.14000
2.00000	14.28000
3.50000	25.00000
7.00000	50.00000

✓ Make videos III: Specify names, images, and speeds

Wrote char2.png
Assignment: place your characters

1. ENTER the names of the characters for your video.
2. ENTER filenames for each character in your video.
3. ENTER the speeds for your characters.

x (seconds)	y (feet)
-------------	----------

```

1 # Enter the names of your characters:
2 names = ["Race Car", "Curious George", "doggy", "low-bike"]
3
4 # Enter the file names of your images:
5 images = ["race-car.jpg", "Curious-George.jpeg", "dog.jpg", "low-bike.png"]
6
7 # Enter the speeds:
8 speeds = [12.5, 3.84, 7.14, 6.25 ]
9
10
11 # Directory:

```



```

12 directory = "/content/drive/MyDrive/Race-2025/GraphSpeeds2-main/"
13 images_list = [directory + filename for filename in images]
14
15 # Save all of the data:
16 data = [names, images_list, speeds]

```

✓ Make videos IV: Make and experiment with your video

Wrote char3.png

Your video will be stored to the video file race.mp4 or a filename that you choose.

To make a video, one time unit (e.g., one hour) corresponds to one second of video.

If you set the simulation_speed=2.0, then one second will correspond to two time units (e.g., two hours).

For each second, the video contains a number of images. This is controlled by fps which means number of frames per second.

x (seconds)	y (feet)
0.00000	0.00000
1.00000	6.25000
2.00000	12.50000
4.00000	25.00000
8.00000	50.00000

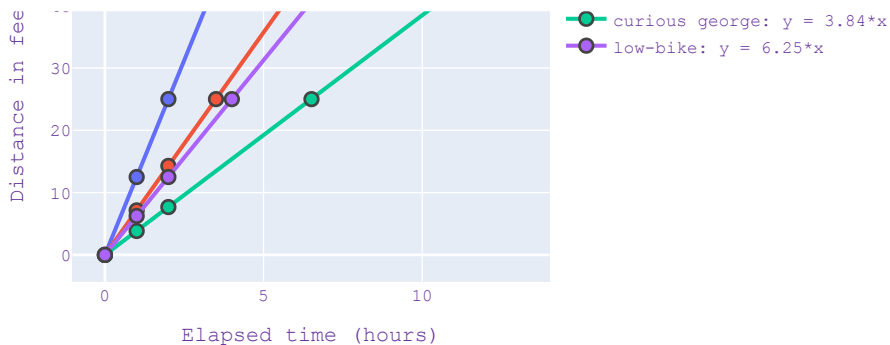
Assignment: experiment with your video

1. Enter the race distance, duration of the race, and video title.
2. Enter the units.
3. Change simulation_speed to 2.0. What do you observe?
4. Change fps to 20. What do you observe?
5. Save the video to a different file name.

```

1 # Experiment with the video
2 sim_video = simulationVideo(data = data,          # characters
3                             duration = 14,        # ENTER duration
4                             race_distance = 50,    # ENTER race distance
5                             vid_title = "Race")    # video title
6
7 sim_video.set_video(video_name="race.mp4",        # new filename
8                     fps=10,                        # frames per second
9                     simulation_speed=1.0)          # speed of the simulation
10
11 sim_video.set_units(distance_string="feet",        # ENTER distance unit.
12                     time_string="seconds",        # ENTER time unit.
13                     speed_string="feet/second")    # ENTER speed units.
14
15
16 # Create and save your video:
17 video = sim_video.create_video()
18 video.ipython_display()

```



```
File 'race.mp4' has been removed.
VideoWriter initialized successfully.
video file = race.mp4 closed.
Moviepy - Building video __temp__.mp4.
Moviepy - Writing video __temp__.mp4

Moviepy - Done !
Moviepy - video ready __temp__.mp4
```

✓ Make videos V: Make title and explanatory text images

You can create many text frames and save them into different frames.

This is done using `saveImage("filename.png")` as given below:

```
text = """"First line.
Second line.
Third line.""
textImage("frame1.png", text)
```

Assignment: Create text images

Fill out the explanatory text images.

```
1 # Create Title and group name here
2 text = """"Session 1 Group D
3 WHO IS THE FASTEST
4 """"
5 textImage("frame1.png", text)
6
7
8 # Who is racing?
9 text = """"Race car, Curious Gorge, doggy, low-bike
10 """"
11 textImage("frame2.png", text)
12
13
14 # Why they are racing?
15 text = """"to see who is fast
16 """"
```

```

17 textImage("frame3.png", text)
18
19
20 # Character 1 equation and explanation:
21 text = """"race car equation
22 y = 12.5*x
23 The race car is going at
24 12.6 feet per seconds ...
25 """"
26 textImage("frame4.png", text)
27
28
29 # Character 2 equation and explanation:
30 text = """"Curious George equation:
31 y = 3.84 * x
32 Mr.Curious George is going
33 3.84 feet per seconds
34 """"
35 textImage("frame5.png", text)
36
37
38 # Character 3 equation and explanation:
39 text = """"Doggy equation:
40 y = 7.14 * x
41 doggy is going
42 7.14 feet per seconds
43 """"
44 textImage("frame6.png", text)
45
46
47 # Character 4 equation and explanation:
48 text = """"low bike equation:
49 y = 6.25 * x
50 Mr. Low Bike is going
51 6.25 feet per second.
52 """"
53 textImage("frame7.png", text)
54
55
56 # Explain the winner:
57 text = """"Race car won because
58 12.5 feet per seconde is the fastest
59 """"
60 textImage("frame8.png", text)
61
62
63 # Victory page:
64 text = """"
65 thanks for watching .
66 """"
67 textImage("frame9.png", text)

```

✓ Make videos VII: Make the final video

You can make the final video by combining images and videos together.

Assignment: Make final video

Verify that your video satisfies the requirements of your assignment!

```

1 # Confirm that you have all the image here:
2 img_list = ["frame1.png", "frame2.png", "frame3.png",
3             "race.mp4",
4             "char1.png", "frame4.png",
5             "char2.png", "frame5.png",
6             "char3.png", "frame6.png",
7             "char4.png", "frame7.png", "race.png",
8             "frame8.png", "frame9.png"]
9
10 # Create the video:
11 video=CreateVideo(video_name="all.mp4",      # This is file name of the final video.
12                   file_list=img_list,        # This the list of images and files to combine.

```

```
13             fps=10,                # The number of frames per second.
14             duration=1.0)          # The duration for each image.
15
16 # Display the video:
17 video.ipython_display()
```

```
Opening frame1.png
Opening frame2.png
Opening frame3.png
Opening race.mp4
Opening char1.png
Opening frame4.png
Opening char2.png
Opening frame5.png
Opening char3.png
Opening frame6.png
Opening char4.png
Opening frame7.png
Opening race.png
Opening frame8.png
Opening frame9.png
video: height = 600 width = 800
File 'all.mp4' has been removed.
VideoWriter initialized successfully.
Moviepy - Building video __temp__.mp4.
Moviepy - Writing video __temp__.mp4

Moviepy - video ready __temp__.mp4
```

Moviepy - Done !

✓ SAVE YOUR VIDEOS AND GOOGLE COLAB NOTEBOOKS

Double-click (or enter) to edit

1 Start coding or [generate](#) with AI.

```
1 from moviepy.editor import concatenate_videoclips
2
3 clips = []
4
5 for i, filename in enumerate(img_list):
6
7     # Give EVERY clip its own unique temporary filename
```

```
8     temp_name = f"temp_{i}.mp4"
9
10    if filename.startswith("char"):
11        duration = 4.0
12
13    elif filename in ["frame4.png", "frame5.png", "frame6.png", "frame7.png"]:
14        duration = 8.0
15
16    elif filename in ["frame8.png"]:
17        duration = 8.0
18
19    elif filename.startswith("frame"):
20        duration = 4.0
21
22    else:
23        duration = 6
24
25
26    clip = CreateVideo(
27        video_name=temp_name,
28        file_list=[filename],
29        fps=10,
30        duration=duration
31    )
32
33    clips.append(clip)
34
35 # Combine everything in the original order
36 final video = concatenate videoclipss(clips. method="compose")
```